

CEN-124 CIRCUIT THEORY

Textbook: Electric Circuits (Seventh Edition)
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Course Content

- Circuit Variables
- Circuit Elements
- Simple Resistive Circuits
- Techniques of Circuit Analysis
- The Operational Amplifier
- Inductance, Capacitance and Mutual Inductance
- Response of First-Order RL and RC Circuits
- Natural and Step Responses of RLC Circuits

Chapter-1 Circuit Variables

- There are three important things in circuit analysis
 - I Definitions (current, voltage, power, circuits, circuit elements, etc.)
 - II Laws: To write equations in terms of current and voltage.
 - III Techniques of circuit analysis (to obtain solution)
- We will work on these areas.
- Circuit theory is a special case of electromagnetic field theory: the study of static and moving electric charges.
- If $\lambda \gg d$ (the wavelength is much greater than the physical dimension of the circuit) then we can use the circuit theory instead of electromagnetic field theory to find currents and voltages. The circuit theory provides simple solutions (of sufficient accuracy) to problems that would become hopelessly complicated if we were to use electromagnetic field theory.

Chapter-1 Circuit Variables

$$\text{If } f = 50 \text{ Hz} \Rightarrow \lambda = \frac{c(\text{speed of light})}{f(\text{frequency})} = \frac{3 \times 10^8 \text{ m/s}}{50 \text{ 1/s}} = 6000 \text{ km}$$

if $d \ll \lambda$ circuit analysis can be employed to obtain a solution otherwise we must use electromagnetic field theory (need to solve Maxwell's equations.)

If $10d < \lambda \Rightarrow$ we assume $d \ll \lambda$

Voltage and Current

- The Concept of electric charge is the basis for describing all electrical phenomena.
- The charge is bipolar, meaning that electrical effects are described in terms of positive and negative charges
- The electric charge exists in discrete quantities, which are integral multiples of the electronic charge, $1.6022 \times 10^{-19} \text{ C}$.
- Electric effects are attributed to both the separation of charge and charges in motion.
- In circuit theory, the separation of charge creates an electric force (voltage), and motion of charge creates an electric fluid(current)
- Voltage is the energy per unit charge created by the separation.

Chapter-1 Circuit Variables

-The rate of charge flow is known as the electric current.

Using the definition of voltage, we can write

$$v = \frac{dw}{dq}$$

where v is the voltage in volts, w is the energy in joules, and q is the charge in coulombs. Using the definition of current, we can write

$$i = \frac{dq}{dt}$$

where i is the current in amperes, t is the time in seconds, and q is the charge in coulombs.

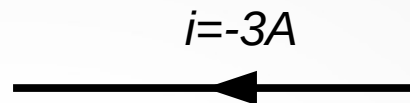
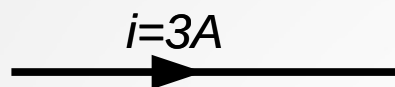
-The power is defined as energy per unit time. We can write

$$p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = v * i$$

where p is the power in watts, t is the time in seconds, w is the energy in joules, i is the current in amperes, and v is the voltage in volts.

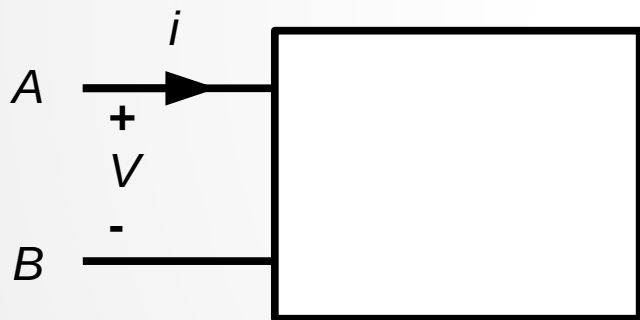
Chapter-1 Circuit Variables

-Two methods of representation for the same current.

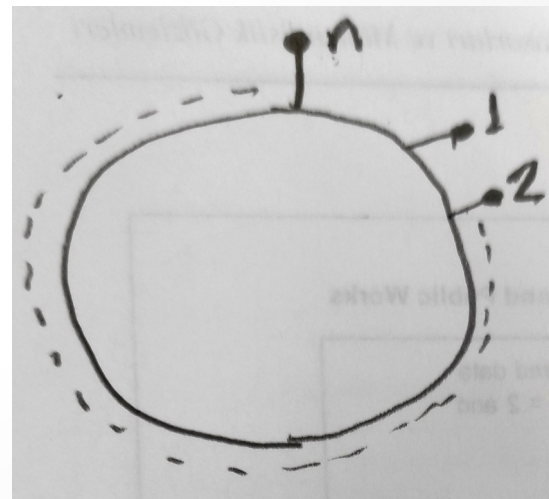


So you should define a reference direction while writing the equations or talking about the current. The same is true for the voltage.

-A general circuit element is characterized by a pair of terminal and other circuit elements are connected to these terminals. It is described in terms of voltage and current.



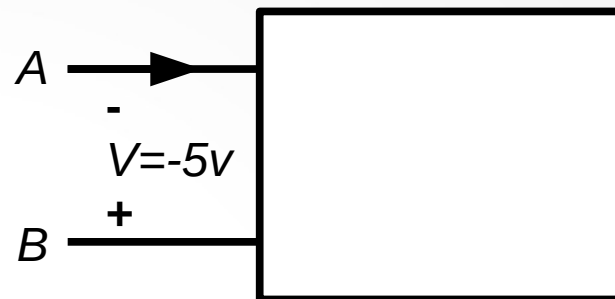
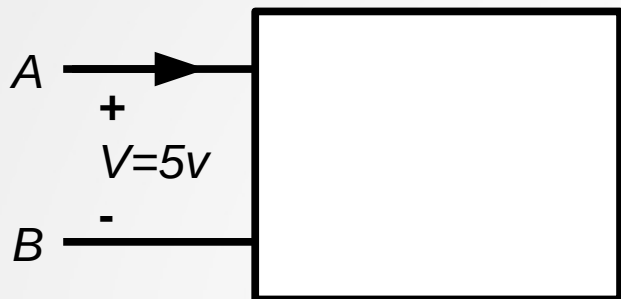
A basic circuit terminals



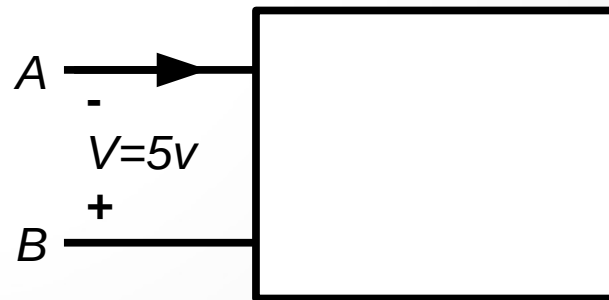
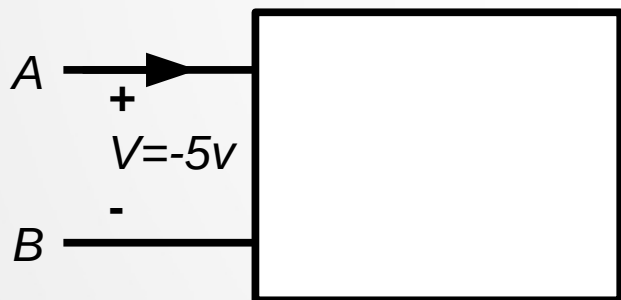
A more complicated circuit element with n terminals

Chapter-1 Circuit Variables

-So as seen for the current, we should define reference direction for the voltage too.



-Terminal A is 5 volt positive with respect to terminal B.

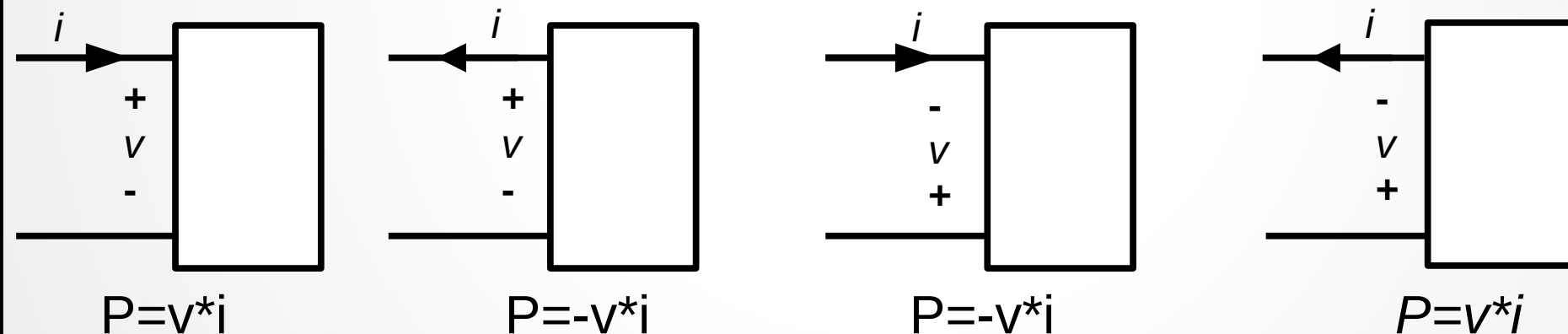


-Terminal B is 5 volt positive with respect to terminal A.

Chapter-1 Circuit Variables

Passive Sign Convention

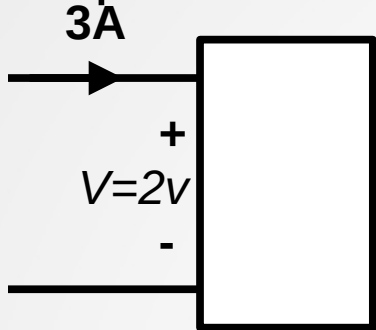
- The assignment of the reference polarity for voltage and reference direction for current and voltage are entirely arbitrary.
- If the reference direction of the current is from the “+” side of the voltage reference through the circuit element to the “-” side of the voltage reference then $P=v*i$. Otherwise $P=-v*i$. (If the current enters from positive side $P=v*i$, otherwise $P=-v*i$)
- If $P>0$, circuit element absorbs (dissipates) power. Otherwise it delivers (generates) power.



Polarity of references and the expression for power.

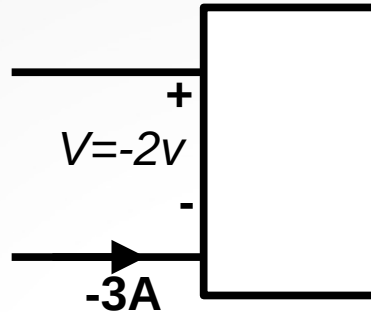
Chapter-2 Circuit Elements

Examples



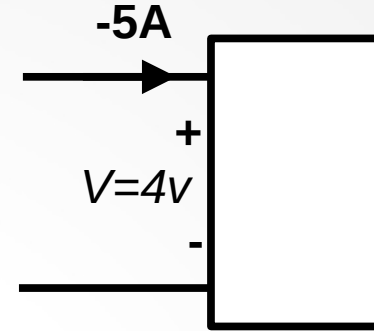
$$P = 3 \times 2 = 6\text{W}$$

Absorbs



$$P = -(-3) \times (-2) = -6\text{W}$$

Generates

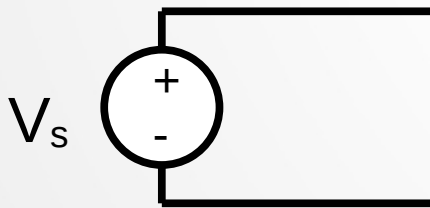


$$P = 4 \times (-5) = -20\text{W}$$

Generates

Voltage and Current Sources

Independent Sources

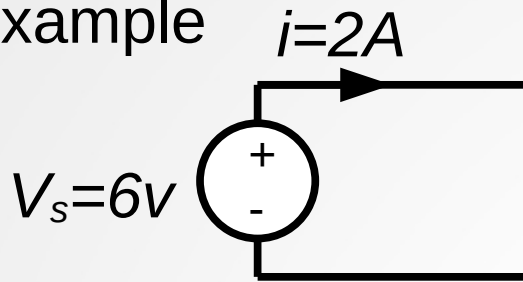


Upper terminal is V_s volt positive
with respect to lower terminal

Independent voltage source

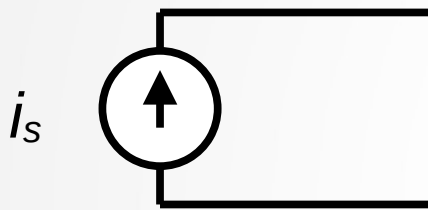
Chapter-2 Circuit Elements

Example



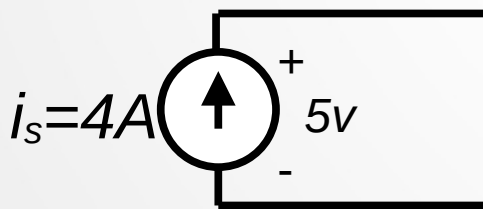
$$P = -(6 \times 2) = -12W$$

The voltage source is delivering $12W$. Sources are expected to deliver power to the network and not to absorb it.



The circuit symbol for an independent current source.

Example

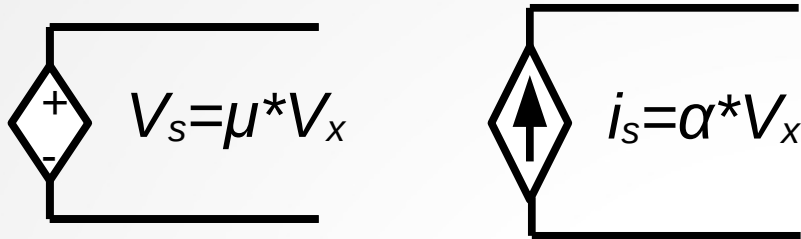


$$P = -4 \times 5 = -20W$$

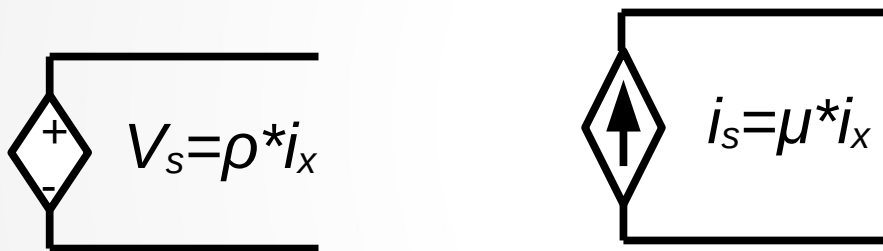
The current source is delivering $20W$.

Chapter-2 Circuit Elements

Dependent Sources



Voltage controlled dependent sources



Current controlled dependent sources

A dependent source establishes a voltage or current whose value depends on the value of a voltage or current on which it depends.

Chapter-2 Circuit Elements

Electrical Resistance (Ohm's law. George Simon Ohm)

Ohm's law states that voltage across many types of materials is directly proportional to the current flowing through the material and can be written as

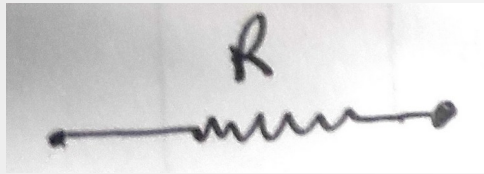
$$V = R \cdot i$$

where R is the resistance in Ohms (V/A, denoted by Ω), V is the voltage in volts, and i is the current in amperes.

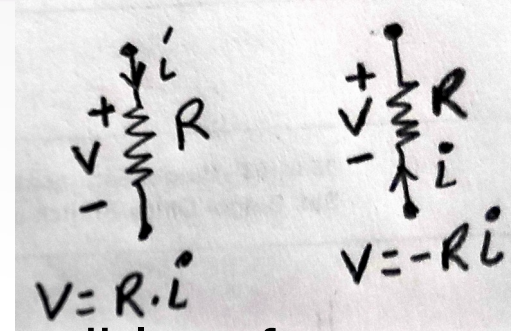
Most materials exhibit measurable resistance to current. The amount of resistance depends on the material. Most metals such as copper and aluminum have small values of resistance, making them good choices for wiring used to conduct electrical current. Many useful electrical devices take advantage of resistance heating, including stoves, toasters, irons, and space heaters.

For the purposes of circuit analysis, we must reference the current in the resistor to the terminal voltage. We can do so in two ways: either in the direction of the voltage drop across the resistor or in the direction of the voltage rise across the resistor, as shown on the next page.

Chapter-2 Circuit Elements



The circuit symbol for a resistor having a resistance R .



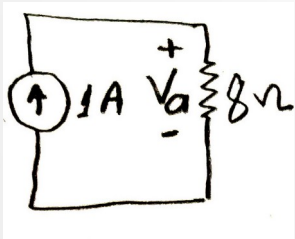
Two possible reference choices for the current and voltage at the terminals of a resistor, and resulting equations.

$$P = v * i = i^2 * R = v^2 / R$$

$$P = -v * i = -(-R * i) * i = i^2 * R = v^2 / R \quad (V = -i * R \text{ case})$$

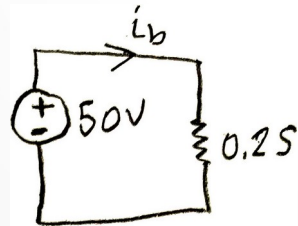
The ratio of current to voltage is also a constant and written as $\frac{i}{v} = G$ is called conductance and denoted by σ (inverted Ω).

The unit is mho or siemens (S).



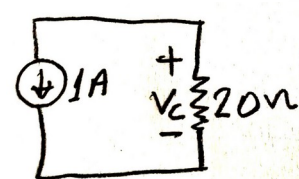
$$V_a = 1 * 8 = 8 \text{ V}$$

$$P = 8 * 1 = 8 \text{ W}$$



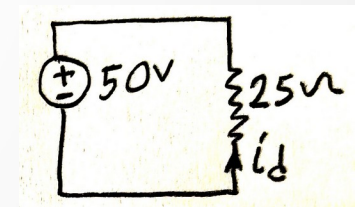
$$i_b = 50 * 0.2 = 10 \text{ A}$$

$$P = 10 * 50 = 500 \text{ W}$$



$$V_c = -1 * 20 = -20 \text{ V}$$

$$P = -(-20) * (1) = 20 \text{ W}$$



$$i_d = -50 / 25 = -2 \text{ A}$$

$$P = -50 * (-2) = 100 \text{ W}$$

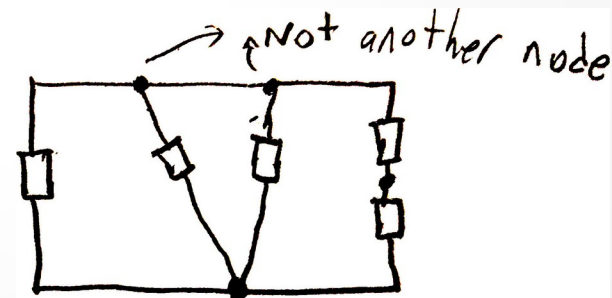
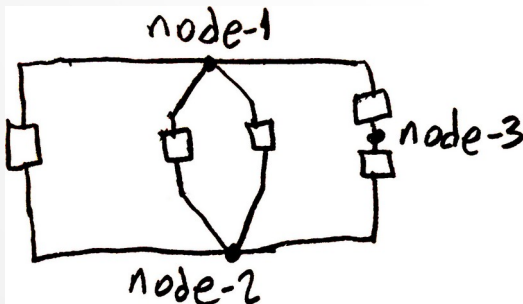
Chapter-2 Circuit Elements

Kirchhoff's Laws

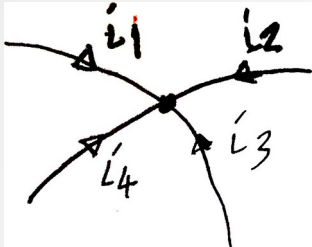
a) Kirchhoff's Current Law (KCL): The algebraic sum of all the currents at any node in a circuit equals zero.

Node: A point at which two or more circuit elements have a common connection is called a node.

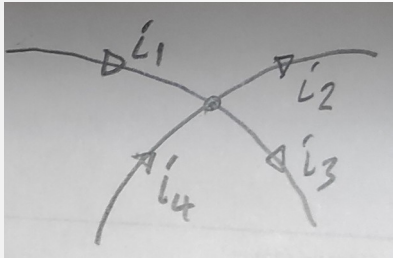
- We may also state Kirchhoff's current law in other ways:
- The algebraic sum of all the currents entering a node in a circuit equals zero.
- The algebraic sum of all the currents leaving a node in a circuit equals zero.
- The sum of all the currents leaving a node in a circuit equals to sum of the currents entering the node.



Chapter-2 Circuit Elements



$$\sum_{k=1}^n i_k = i_1 + i_2 + i_3 + i_4 = 0$$



$$\sum_{k=1}^n i_k = i_1 - i_2 - i_3 + i_4 = 0 \quad \text{Sum of the currents entering the node}$$

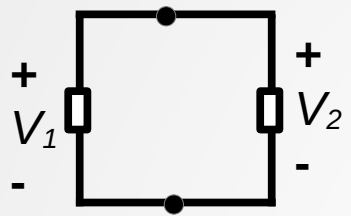
$$\sum_{k=1}^n i_k = -i_1 + i_2 + i_3 - i_4 = 0 \quad \text{Sum of the currents leaving the node}$$

$$i_1 + i_4 = i_2 + i_3 \quad \text{Sum of the currents entering the node} = \text{Sum of the currents leaving the node}$$

-In any circuit with n nodes, $n-1$ independent current equations can be derived from KCL.

Chapter-2 Circuit Elements

b) Kirchhoff's Voltage Law (KVL): The algebraic sum of voltages around any closed path in a circuit equals zero.

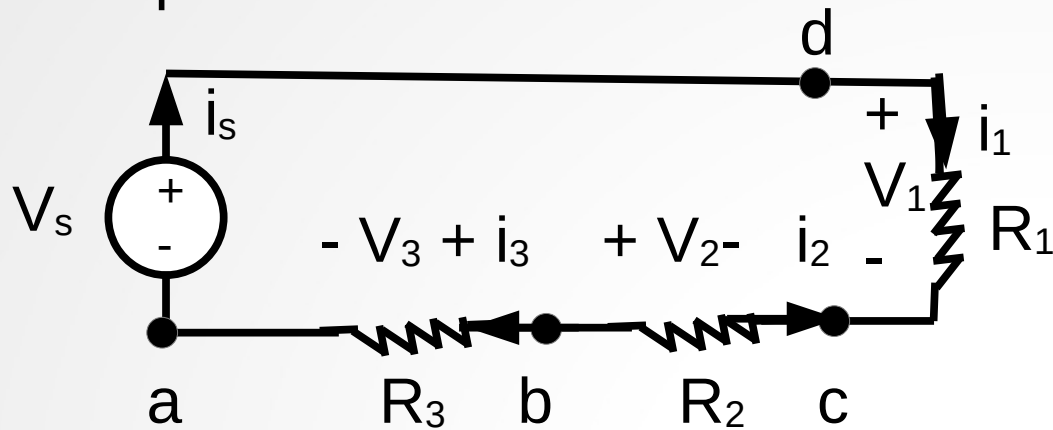


$$V_2 - V_1 = 0 \Rightarrow V_2 = V_1$$

-We must assign an algebraic sign (reference direction) to each voltage in the loop. As we trace a closed path, a voltage will appear either as a rise or a drop in the tracing direction. Assigning a positive sign to a voltage rise requires assigning a negative sign to a voltage drop. Conversely, giving a negative sign to a voltage rise requires giving a positive sign to a voltage drop.

Chapter-2 Circuit Elements

Example



KCL:

Node a $i_s = i_3$

Node b $i_3 = -i_2$

Node c $i_2 = -i_1$

Node d $i_s = i_1$

$$\Rightarrow i_s = i_1 = -i_2 = i_3$$

Ohm's law:

$$V_1 = i_1 * R_1$$

$$V_2 = i_2 * R_2$$

$$V_3 = i_3 * R_3$$

$$i_1 = i_s$$

$$i_2 = -i_s$$

$$i_3 = i_s$$

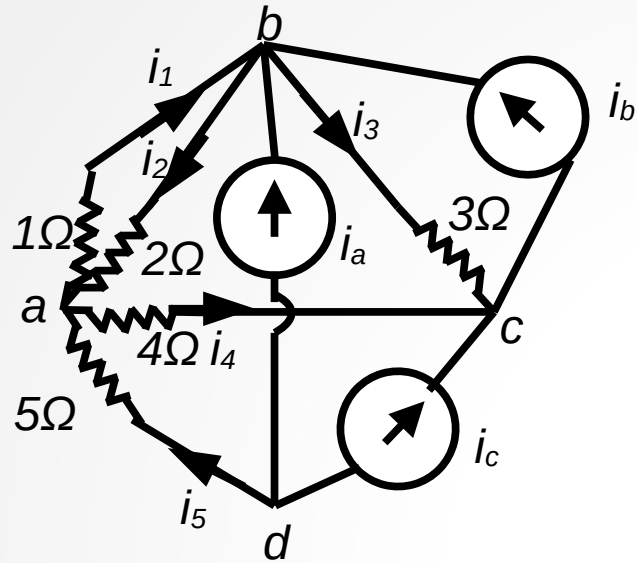
KVL:

$$V_1 - V_2 + V_3 - V_s = 0 \Rightarrow i_1 * R_1 - i_2 * R_2 + i_3 * R_3 - V_s = 0$$

$$\Rightarrow i_s * R_1 + i_s * R_2 + i_s * R_3 - V_s = 0$$

Chapter-2 Circuit Elements

Example: Using Kirchhoff's current law



KCL for

Node a: $i_1 - i_2 + i_4 - i_5 = 0$

Node b: $-i_1 + i_2 + i_3 - i_a - i_b = 0$

Node c: $-i_3 - i_4 + i_b - i_c = 0$

Node d: $i_5 + i_a + i_c = 0$

Chapter-2 Circuit Elements

Example: Using Kirchhoff's voltage law

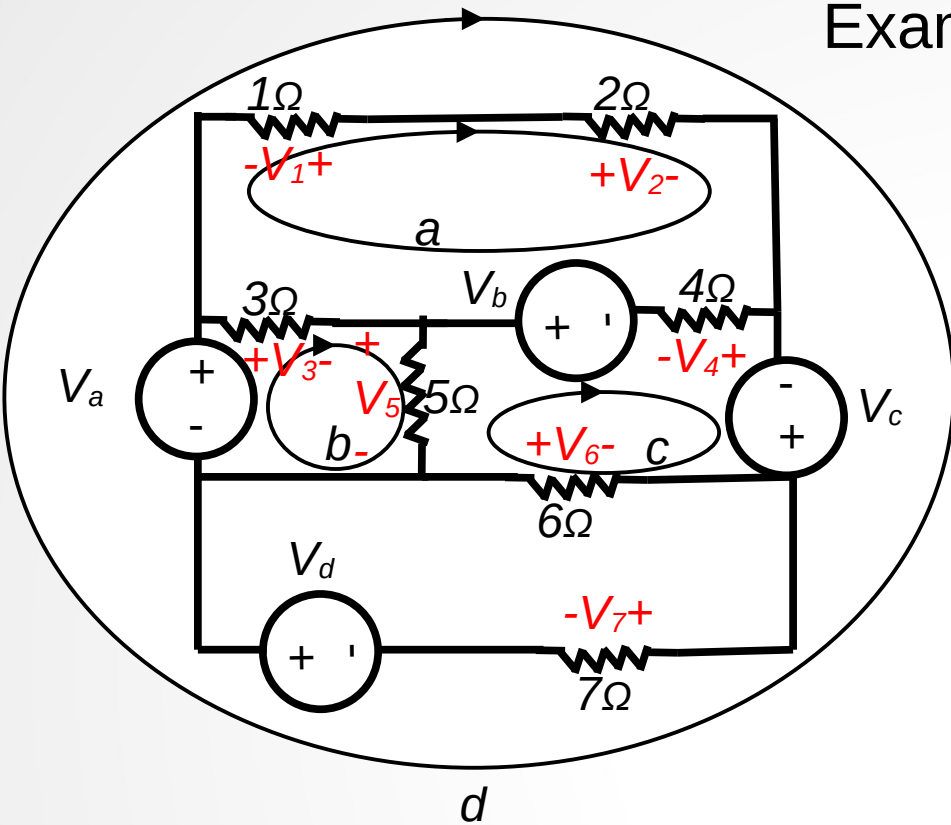
KVL for

Loop a: $-V_1 + V_2 + V_4 - V_b - V_3 = 0$

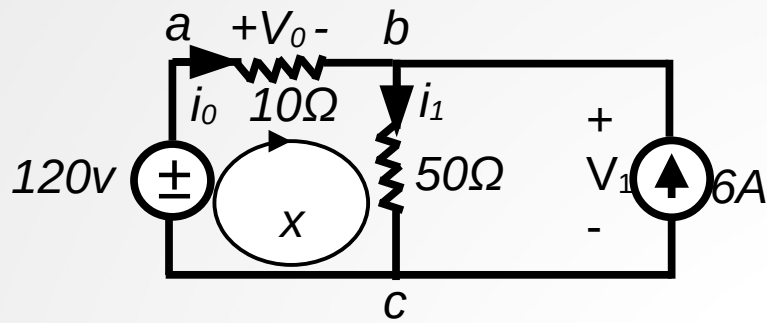
Loop b: $-V_a + V_3 + V_5 = 0$

Loop c: $V_b - V_4 - V_c - V_6 - V_5 = 0$

Loop d: $-V_a - V_1 + V_2 - V_c + V_7 - V_d = 0$



Chapter-2 Circuit Elements



a) Use Kirchhoff's law and Ohm's law to find i_0 in the circuit.

b) Test the solution for i_0 by verifying that the total power generated equals to the total power dissipated.

a) KCL for the node b

$$i_1 - i_0 - 6 = 0 \Rightarrow i_0 = i_1 - 6$$

KVL for loop-x

$$V_0 + V_1 - 120 = 0$$

Ohm's law

$$V_0 = 10i_0$$

$$V_1 = 50i_1$$

$$\Rightarrow 10(i_1 - 6) + 50i_1 = 120$$

$$\Rightarrow 60i_1 = 180 \Rightarrow i_1 = 3 \text{ A}$$

$$i_0 = i_1 - 6 = -3 \text{ A}$$

$$V_1 = 50i_1 = 50 \times 3 = 150 \text{ V}$$

b) Power dissipated in the resistors

$$P_{50\Omega} = 3^2 \times 50 = 450 \text{ W}$$

$$P_{10\Omega} = (-3)^2 \times 10 = 90 \text{ W}$$

Power delivered by the sources:

$$P_{120\text{V}} = -120 \times i_0 = -120 \times (-3) = 360 \text{ W (dissipates)}$$

$$P_{6\text{A}} = -V_1 \times 6 = -150 \times 6 = -900 \text{ W (generates)}$$

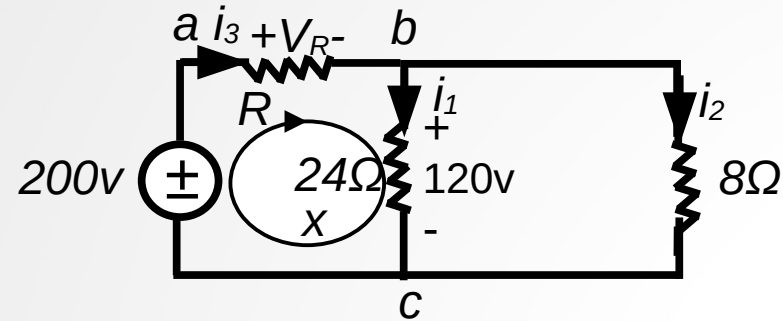
Generated power = 900W by the current source

Dissipated power = 450 + 90 + 360 = 900W

$\Rightarrow$ generated power = 900W = dissipated power

Chapter-2 Circuit Elements

Example: Use Ohm's law and Kirchhoff's laws to find the value of R in the circuit shown below.



Using Ohm's law

$$i_1 = \frac{120}{24} = 5 \text{ A}$$

$$i_2 = \frac{120}{8} = 15 \text{ A}$$

Using KCL for node b

$$i_3 = i_1 + i_2 = 5 + 15 = 20 \text{ A}$$

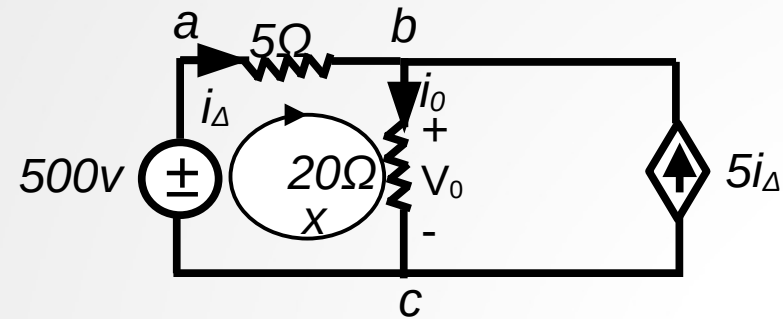
Using Ohm's law and KVL for loop X

$$-200 + R * i_3 + 120 = 0 \Rightarrow R = \frac{80}{i_3} = \frac{80}{20} = 4 \Omega$$

Chapter-2 Circuit Elements

Analysis of a Circuit Containing Dependent Sources

Example:



Using KCL for node b: $i_0 = i_{\Delta} + 5i_{\Delta} = 6i_{\Delta}$

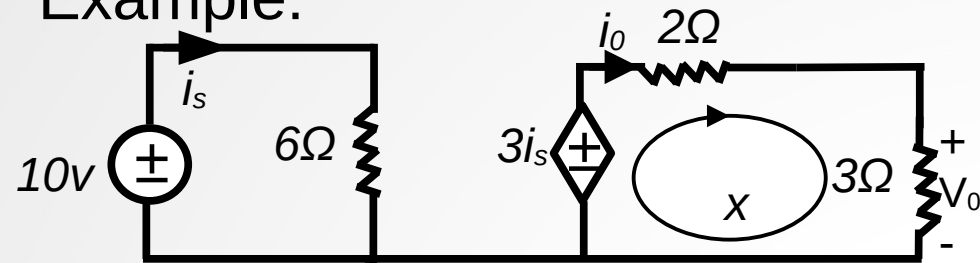
Using KVL for loop-X: $-500 + 5i_{\Delta} + 20i_0 = 0$

$$\Rightarrow 5i_{\Delta} + 20 * (6i_{\Delta}) = 500 \Rightarrow i_{\Delta} = \frac{500}{125} = 4 \text{ A}$$

$$i_0 = 6i_{\Delta} = 24 \text{ A}, \quad V_0 = 20i_0 = 480 \text{ V}$$

Chapter-2 Circuit Elements

Example:



a) Find V_0

b) Show that the total power generated in the circuit equals to the total power dissipated.

Using Ohm's law: $i_s = \frac{10}{6} \text{ A}$

Using Ohm's law and KVL for loop-X:
 $-3i_s + 2i_0 + 3i_0 = 0$

$$\Rightarrow i_0 = \frac{3i_s}{5} = 1 \text{ A}$$

Using Ohm's law: $V_0 = 3i_0 = 3 \text{ V}$

Powers dissipated in the resistors:

$$P_{6\Omega} = 6 * i_s^2 = 6 * \left(\frac{10}{6}\right)^2 = \frac{100}{6} \text{ Watt}$$

$$P_{2\Omega} = 2 * i_0^2 = 2 * 1^2 = 2 \text{ Watt}$$

$$P_{3\Omega} = 3 * i_0^2 = 3 * 1^2 = 3 \text{ Watt}$$

Powers delivered by the sources:

$$P_{10V} = -10 * i_s = -\frac{100}{6} \text{ W}$$

$$P_{3i_s} = -3i_s * i_0 = -3 * \frac{10}{6} * 1 = -5 \text{ W}$$

The total power generated:

$$\frac{100}{6} + 5 = \frac{130}{6} \text{ W}$$

The total power dissipated:

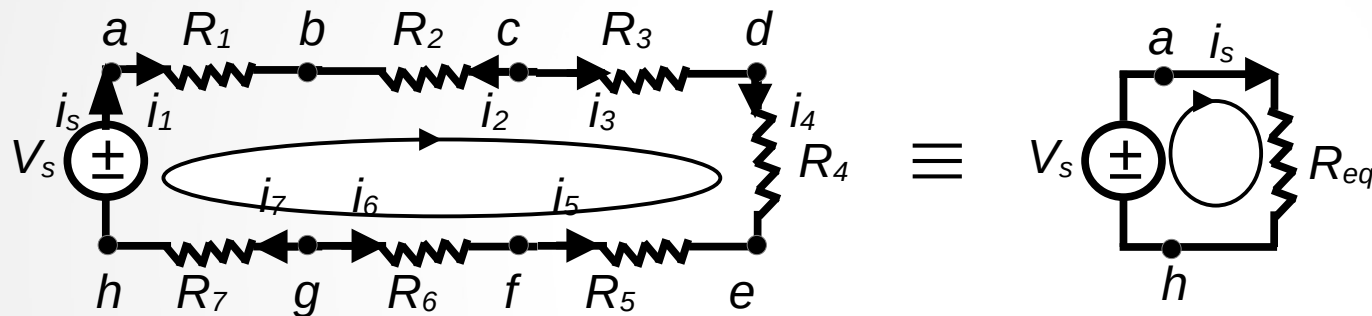
$$2 + 3 + \frac{100}{6} = \frac{130}{6} \text{ W}$$

Chapter-3 Simple Resistive Circuits

Simple Resistive Circuits

Resistors in Series

- When just two circuit elements connect at a single node, they are said to be in series.
- Series-connected circuit elements carry the same current.



KVL for the loop

$$-V_s + i_s * R_{eq} = 0$$
$$\Rightarrow V_s = i_s * R_{eq}$$

Using KCL for nodes a, b, c, d, e, f, and h we get

$$i_s = i_1 = -i_2 = i_3 = i_4 = -i_5 = -i_6 = i_7$$

If there are only two circuit elements connected to the node, we do not need to use KCL.

Using KVL for the loop:

$$-V_s + i_s * R_1 + i_s * R_2 + i_s * R_3 + i_s * R_4 + i_s * R_5 + i_s * R_6 + i_s * R_7 = 0$$

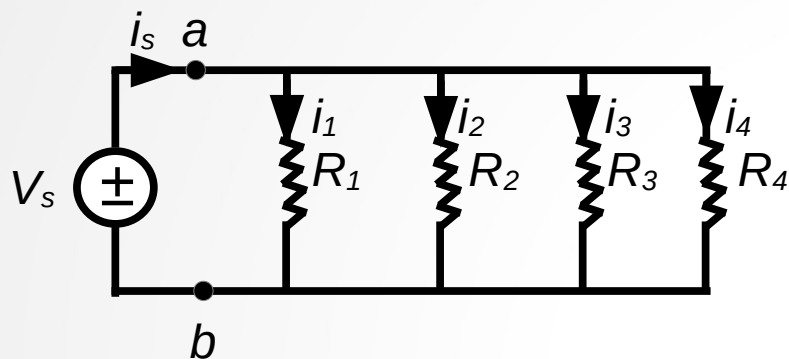
$$\Rightarrow V_s = i_s * (R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7) = i_s * R_{eq}$$

$$\Rightarrow R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7$$

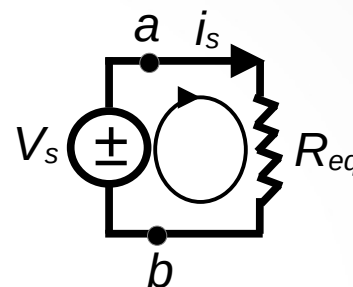
Chapter-3 Simple Resistive Circuits

Resistors in Parallel

-When two circuit elements connect at a single node pair, they are said to be in parallel.



$\equiv$



KVL for the loop

$$-V_s + i_s * R_{eq} = 0$$

$$\Rightarrow V_s = i_s * R_{eq}$$

Resistors in parallel

Using KCL for node a

$$i_1 + i_2 + i_3 + i_4 = i_s$$

Using Ohm's law

$$i_1 = \frac{V_s}{R_1}, \quad i_2 = \frac{V_s}{R_2}, \quad i_3 = \frac{V_s}{R_3}, \quad i_4 = \frac{V_s}{R_4},$$

$$\Rightarrow i_s = V_s * \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$$

$$\Rightarrow i_s = \frac{V_s}{R_{eq}} = V_s \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right)$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$$

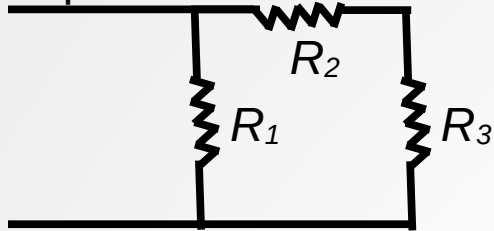
If we use conductance instead of resistance

$$G_1 = \frac{1}{R_1}, \quad G_2 = \frac{1}{R_2}, \quad G_3 = \frac{1}{R_3}, \quad G_4 = \frac{1}{R_4}$$

$$\Rightarrow G_{eq} = G_1 + G_2 + G_3 + G_4$$

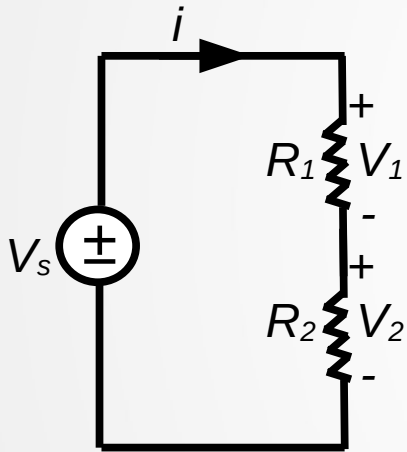
Chapter-3 Simple Resistive Circuits

Example



Non parallel resistors

The Voltage Divider Circuit



Designed to divide the voltage

Using KVL

$$-V_s + i * R_1 + i * R_2 = 0$$

$$\Rightarrow i = \frac{V_s}{R_1 + R_2}$$

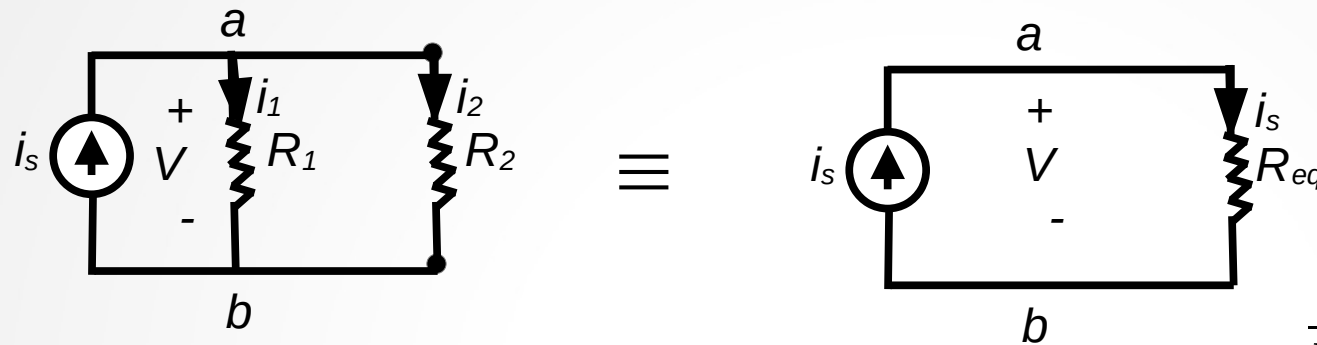
Using Ohm's law

$$V_1 = R_1 * i = V_s * \left(\frac{R_1}{R_1 + R_2} \right)$$

$$V_2 = R_2 * i = V_s * \left(\frac{R_2}{R_1 + R_2} \right)$$

Chapter-3 Simple Resistive Circuits

The Current Divider Circuit -Designed to divide the current



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_{eq} = \frac{R_1 * R_2}{R_1 + R_2}$$

Using Ohm's law

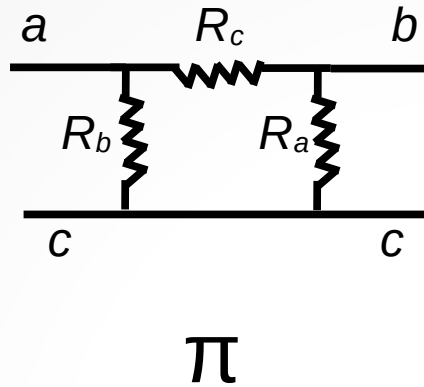
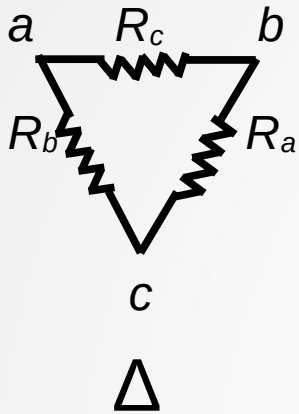
$$V = i_1 * R_1 = i_2 * R_2 = R_{eq} * i_s \quad \left(R_1 \text{ and } R_2 \text{ are connected in parallel } R_{eq} = \frac{R_1 * R_2}{R_1 + R_2} \right)$$

$$i_1 = \frac{V}{R_1} = i_s * \frac{R_2}{R_1 + R_2}$$

$$i_2 = \frac{V}{R_2} = i_s * \frac{R_1}{R_1 + R_2}$$

Chapter-3 Simple Resistive Circuits

Delta-to-Wye (Pi-to-Tee) Equivalent Circuits



$$R_{ab} = \frac{R_c * (R_a + R_b)}{R_a + R_b + R_c} = R_1 + R_2$$

$$R_{bc} = \frac{R_a * (R_b + R_c)}{R_a + R_b + R_c} = R_2 + R_3$$

$$R_{ac} = \frac{R_b * (R_a + R_c)}{R_a + R_b + R_c} = R_1 + R_3$$

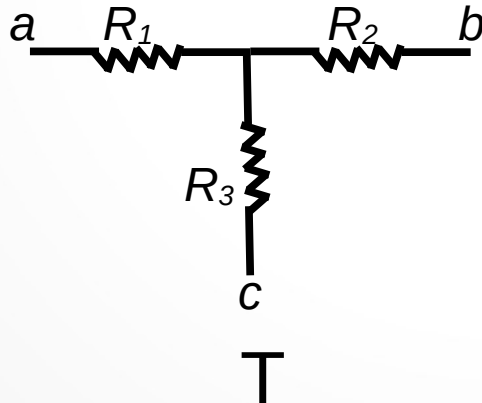
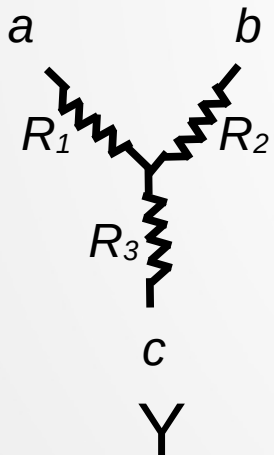
$$R_1 = \frac{R_b * R_c}{R_a + R_b + R_c}, \quad R_2 = \frac{R_a * R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a * R_b}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1 * R_2 + R_2 * R_3 + R_3 * R_1}{R_1}$$

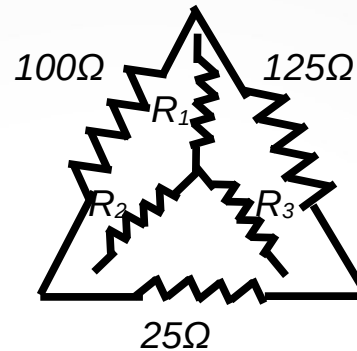
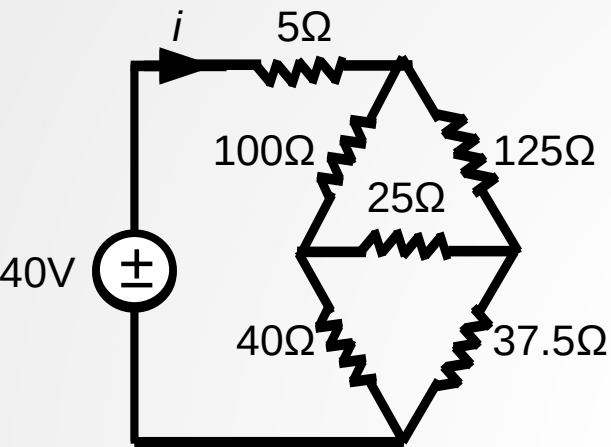
$$R_b = \frac{R_1 * R_2 + R_2 * R_3 + R_3 * R_1}{R_2}$$

$$R_c = \frac{R_1 * R_2 + R_2 * R_3 + R_3 * R_1}{R_3}$$



Chapter-3 Simple Resistive Circuits

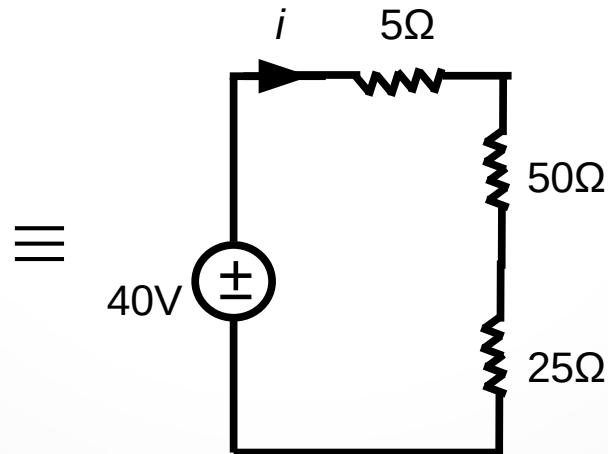
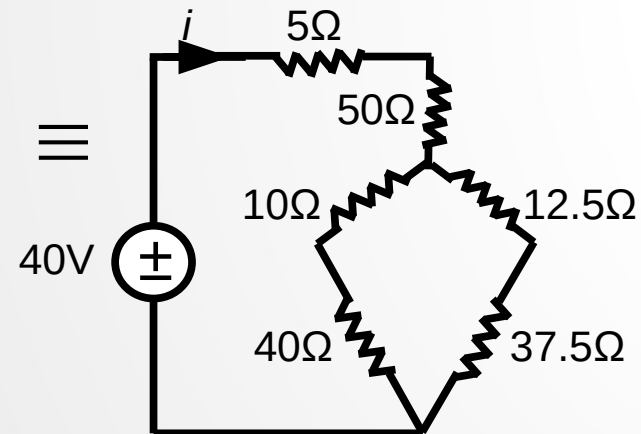
Example: $i = ?$



$$R_1 = \frac{100 * 125}{100 + 125 + 25} = 50 \Omega$$

$$R_2 = \frac{100 * 25}{100 + 125 + 25} = 10 \Omega$$

$$R_3 = \frac{125 * 25}{100 + 125 + 25} = 12.5 \Omega$$



$$i = \frac{40}{80} = 0.5 \text{ A}$$